

Convert $-\frac{13\pi}{5}$ to degrees.

Solution

$$-\frac{13\pi}{5} \times \frac{180^\circ}{\pi}$$
$$-468^\circ$$

1 mark

Question 2

P2, P3

- a) From a group of 9 people, in how many ways can you select a committee of 4 members?
- b) From a group of 9 people, in how many ways can you select a president, a vice president, a secretary, and a treasurer?
- c) Explain why the answers in a) and b) are different.

Solution

a) ${}_9C_4 = 126$ ways

1 mark for ${}_9C_4$

1 mark

b) ${}_9P_4 = 3024$ ways

1 mark for ${}_9P_4$

1 mark

- c) Part a) is a combination because the order does not matter; part b) is a permutation because committee members have specific roles.

1 mark

Question 3

R10

A population of 500 bacteria will triple in 20 hours.

Using the formula given below,

$$A = Pe^{rt}$$

A = population after t hours

P = initial population

r = rate of growth

t = time in hours

- Determine the rate of growth, r .
- Determine how many hours it will take for the initial population to double with the same rate of growth.

Solution

a)

$$1500 = 500e^{r(20)}$$

½ mark for substitution

$$3 = e^{20r}$$

$$\ln 3 = \ln e^{20r}$$

½ mark for applying logarithms

$$\ln 3 = 20r \cdot \ln e$$

½ mark for power rule

$$r = \frac{\ln 3}{20}$$

$$r = 0.054\ 930\ 614$$

½ mark for evaluating quotient of logarithms

2 marks

b)

$$1000 = 500e^{0.054\ 930\ 614t}$$

½ mark for substitution

$$2 = e^{0.054\ 930\ 614t}$$

$$\ln 2 = \ln e^{0.054\ 930\ 614t}$$

½ mark for applying logarithms

$$\ln 2 = 0.054\ 930\ 614t \cdot \ln e$$

½ mark for power rule

$$t = \frac{\ln 2}{0.054\ 930\ 614}$$

$$t = 12.619 \text{ hours}$$

½ mark for evaluating quotient of logarithms

2 marks

Question 4

T5

Talla incorrectly solved the following trigonometric equation:

Solve: $2 \sec x - 5 = 0$; $0^\circ \leq x \leq 360^\circ$.

Talla's work:

$$2 \sec x - 5 = 0$$

$$\sec x = \frac{5}{2}$$

No solution, $\sec x$ cannot be greater than 1.

- Explain her error.
- Determine the correct solution.

Solution

- Talla incorrectly stated that $\sec x$ cannot be greater than 1.
The value of $\cos x$ cannot be greater than 1.

1 mark

$$\text{b) } \sec x = \frac{5}{2}$$

$$\cos x = \frac{2}{5}$$

$$x_r = 66.421\ 821$$

$$x = 66.422^\circ$$

$$x = 293.578^\circ$$

1 mark for reciprocal

1 mark for solving for x ($\frac{1}{2}$ mark for each value of x)

2 marks

Question 5

P4

Simplify the 6th term in the expansion of:

$$\left(2x - \frac{3}{x^2}\right)^{10}$$

Solution

$$t_6 = {}_{10}C_5 (2x)^5 \left(-\frac{3}{x^2}\right)^5$$

2 marks (1 mark for ${}_{10}C_5$, $\frac{1}{2}$ mark for each consistent factor)

$$= 252 \left(32x^5\right) \left(-\frac{243}{x^{10}}\right)$$

$$= -1\,959\,552\,x^{-5}$$

1 mark for simplification ($\frac{1}{2}$ mark for coefficient, $\frac{1}{2}$ mark for exponent)

3 marks

Question 6

T1

Determine the arc length subtended by a central angle if the diameter is 19 cm and the central angle is 1.6 radians.

Solution

$$s = \theta r$$

$$s = (1.6)(9.5)$$

$$s = 15.2 \text{ cm}$$

1 mark

Question 7

T3, T5

Solve the following equation algebraically for x , where $0 \leq x \leq 2\pi$.

$$2 \cos^2 x = -3 \sin x$$

Solution

$$2(1 - \sin^2 x) = -3 \sin x$$

1 mark for identity

$$2 - 2 \sin^2 x = -3 \sin x$$

$$0 = 2 \sin^2 x - 3 \sin x - 2$$

$$0 = (2 \sin x + 1)(\sin x - 2)$$

$$\sin x = -\frac{1}{2}$$

~~$$\sin x = 2$$~~

1 mark for solving for $\sin x$

No Solution

1 mark for indicating no solution

$$x = \frac{7\pi}{6}$$

$$x = \frac{11\pi}{6}$$

1 mark for solving for x ($\frac{1}{2}$ mark for each value)

4 marks

In how many different ways can you arrange the letters in the word VOLLEYBALL?

State your answer as a factorial.

Solution

$$\frac{10!}{4!}$$

1 mark

Note(s):

- award full marks for 151 200

Question 9

R11

Is $(x - 2)$ a factor of the polynomial $p(x) = -x^4 - 3x^3 + 11x^2 + 3x - 10$?

Justify your response.

Solution

Method 1

$$\begin{aligned} p(2) &= -(2)^4 - 3(2)^3 + 11(2)^2 + 3(2) - 10 \\ &= -16 - 24 + 44 + 6 - 10 \\ &= 0 \end{aligned}$$

The remainder is zero, so $(x - 2)$ is a factor.

$\frac{1}{2}$ mark for $p(2)$

1 mark for the remainder theorem

$\frac{1}{2}$ mark for justification

2 marks

Method 2

2	-1	-3	11	3	-10
	↓	-2	-10	2	10
	-1	-5	1	5	<u>0</u>

The remainder is zero, so $(x - 2)$ is a factor.

$\frac{1}{2}$ mark for $x = 2$

1 mark for synthetic division
(or for any equivalent strategy)

$\frac{1}{2}$ mark for justification

2 marks

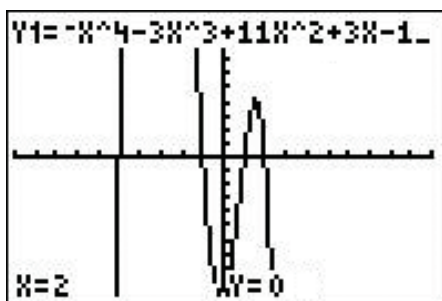
Method 3

I entered $y = -x^4 - 3x^3 + 11x^2 + 3x - 10$ into my calculator and located the zeroes.

$x = 2$ was a zero, which means $(x - 2)$ is a factor.

1 mark for graphing calculator method

1 mark for relating the zeroes to the factors



2 marks

Determine the period of the sinusoidal function $y = \frac{1}{2} \sin\left(\frac{1}{3}x\right)$.

State your answer in radians.

Solution

$$p = \frac{2\pi}{|b|}$$

$$p = \frac{2\pi}{\left|\frac{1}{3}\right|}$$

$$p = 6\pi \quad \text{or} \quad p = 18.850$$

$\frac{1}{2}$ mark for correct value of b

$\frac{1}{2}$ mark for period consistent with b

1 mark

Question 11

R1

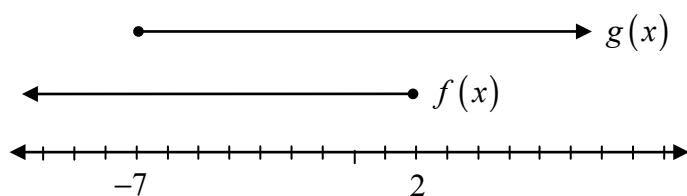
The domain of $f(x)$ is $x \leq 2$. The domain of $g(x)$ is $x \geq -7$.

State the domain of $f(x) + g(x)$.

Justify your answer.

Solution

Both $f(x)$ and $g(x)$ have restricted domains, so both domains need to be considered.



The solution is the intersection of the two domains.

1 mark for justification

$$\{x | x \in \mathbb{R}, -7 \leq x \leq 2\} \quad \text{or} \quad [-7, 2]$$

1 mark for domain

2 marks

Prove the identity below for all permissible values of θ .

$$\frac{1}{1 + \cos \theta} = \csc^2 \theta - \frac{\cot \theta}{\sin \theta}$$

Solution

Method 1

Left-Hand Side	Right-Hand Side	
$\frac{1}{1 + \cos \theta}$	$\csc^2 \theta - \frac{\cot \theta}{\sin \theta}$	
	$\frac{1}{\sin^2 \theta} - \frac{\frac{\cos \theta}{\sin \theta}}{\sin \theta}$	1 mark for correct substitution of identities
	$\frac{1}{\sin^2 \theta} - \frac{\cos \theta}{\sin \theta} \cdot \frac{1}{\sin \theta}$	
	$\frac{1}{\sin^2 \theta} - \frac{\cos \theta}{\sin^2 \theta}$	1 mark for algebraic strategies
	$\frac{1 - \cos \theta}{\sin^2 \theta}$	
	$\frac{1 - \cos \theta}{1 - \cos^2 \theta}$	1 mark for logical process to prove an identity
	$\frac{1 - \cancel{\cos \theta}}{(1 - \cancel{\cos \theta})(1 + \cos \theta)}$	
	$\frac{1}{1 + \cos \theta}$	3 marks

Solution**Method 2**

Left-Hand Side	Right-Hand Side	
$\frac{1}{1 + \cos \theta} (1 - \cos \theta)$	$\csc^2 \theta - \frac{\cot \theta}{\sin \theta}$	1 mark for algebraic strategies
$\frac{1 - \cos \theta}{1 - \cos^2 \theta}$		
$\frac{1 - \cos \theta}{\sin^2 \theta}$		1 mark for correct substitution of identities
$\frac{1}{\sin^2 \theta} - \frac{\cos \theta}{\sin^2 \theta}$		
$\csc^2 \theta - \frac{\cos \theta}{\sin \theta} \cdot \frac{1}{\sin \theta}$		1 mark for logical process to prove an identity
$\csc^2 \theta - \frac{\cot \theta}{\sin \theta}$		3 marks

Explain how the end behaviours of the graphs of polynomial functions with an even degree and with an odd degree are different.

Solution

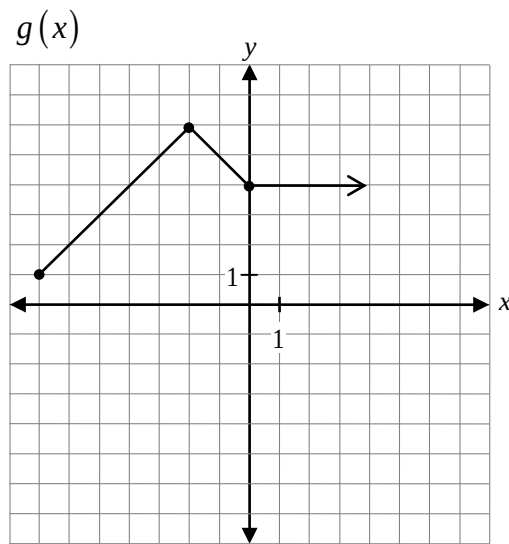
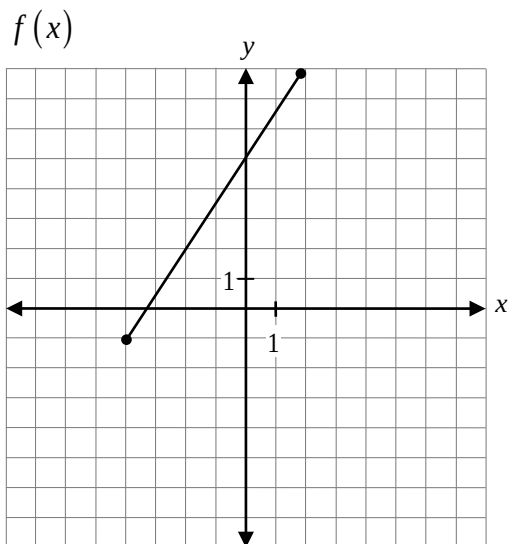
When the degree is odd, the end behaviour is in opposite directions.
When the degree is even, the end behaviour is in the same direction.

1 mark

Question 14

R1

Given the graphs of $f(x)$ and $g(x)$, sketch the graph of $g(x) - f(x)$.



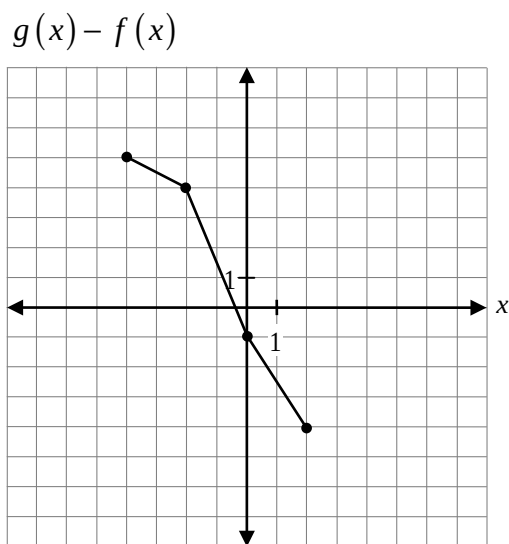
Solution

x	$g(x)$	$f(x)$	$(g - f)(x)$
-4	4	-1	5
-2	6	2	4
0	4	5	-1
2	4	8	-4

1 mark for subtraction of $g(x) - f(x)$

1 mark for restricting domain on graph

2 marks



Given $f(x) = -3x + 7$, evaluate $f^{-1}(-2)$.

Solution

$$\text{Let } y = f(x)$$

$$f(x) = -3x + 7$$

$$y = -3x + 7$$

$$x = -3y + 7$$

$$x - 7 = -3y$$

$$y = \frac{x - 7}{-3}$$

$$f^{-1}(x) = \frac{x - 7}{-3}$$

$$f^{-1}(-2) = \frac{-2 - 7}{-3}$$

$$f^{-1}(-2) = 3$$

1 mark for switching x and y

$\frac{1}{2}$ mark for $f^{-1}(x)$

$\frac{1}{2}$ mark for $f^{-1}(-2)$

2 marks

Question 16

P4

How many terms are there in the expansion of $(x^{12} + 3)^{10}$?

- a) 9
- b) 10
- c) 11
- d) 12

Question 17

T1

A co-terminal angle for $\theta = \frac{11\pi}{3}$ in the domain $-2\pi \leq \theta \leq 0$ would be:

- a) $-\frac{5\pi}{3}$
- b) $-\frac{\pi}{3}$
- c) $\frac{\pi}{3}$
- d) $\frac{5\pi}{3}$

Question 18

R9

The x -intercept of the graph of $y = 3^x - 1$ is:

- a) -1
- b) 0
- c) 1
- d) 2

Question 19

P3

If ${}_nC_5 = {}_nC_3$, the value of n must be:

- a) 3
- b) 5
- c) 8
- d) 15

Question 20

R13

What is the domain of the function $f(x) = \sqrt{-(x+1)}$?

- a) $\{x | x \in \mathbb{R}, x \neq -1\}$
- b) $\{x | x \in \mathbb{R}, x \geq -1\}$
- c) $\{x | x \in \mathbb{R}, x \leq -1\}$
- d) $\{x | x \in \mathbb{R}\}$

Question 21

T6

Identify a non-permissible value of x for the expression $\frac{1}{\cos 2x}$.

- a) 0
- b) $\frac{\pi}{4}$
- c) $\frac{\pi}{2}$
- d) π

Question 22

R8

The expression $2 \log x - \frac{1}{3} \log y$ as a single logarithm is:

a) $\log \frac{x^2}{\sqrt[3]{y}}$

b) $\log \frac{2x}{3y}$

c) $-\log x^2 \sqrt[3]{y}$

d) $\log(x^2 - \sqrt[3]{y})$

Question 23

T2

The point $P(\theta)$ lies on the unit circle. What are the coordinates of the point if $\theta = 300^\circ$?

a) $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

b) $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

c) $\left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$

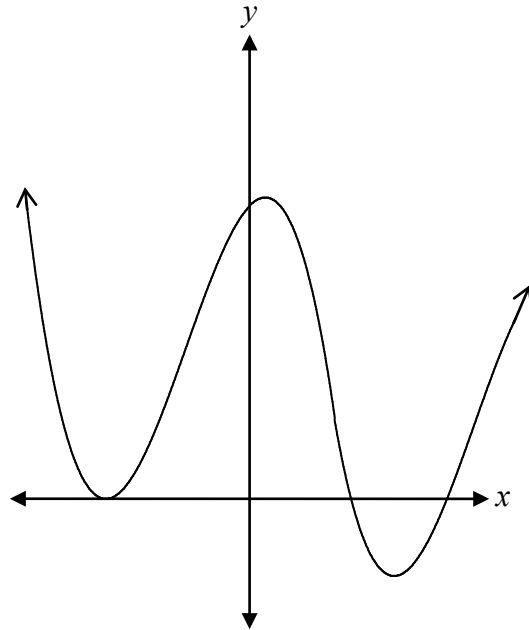
d) $\left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$

Question 24

R12

What is the degree of the polynomial function represented by the graph below?

- a) 2
- b) 3
- c) 4
- d) 5



Question 25

R5

When the point $(-4, -3)$ is reflected over the line $y = x$, the coordinates of the new point are:

- a) $(-3, -4)$
- b) $(3, 4)$
- c) $(4, -3)$
- d) $(-4, 3)$

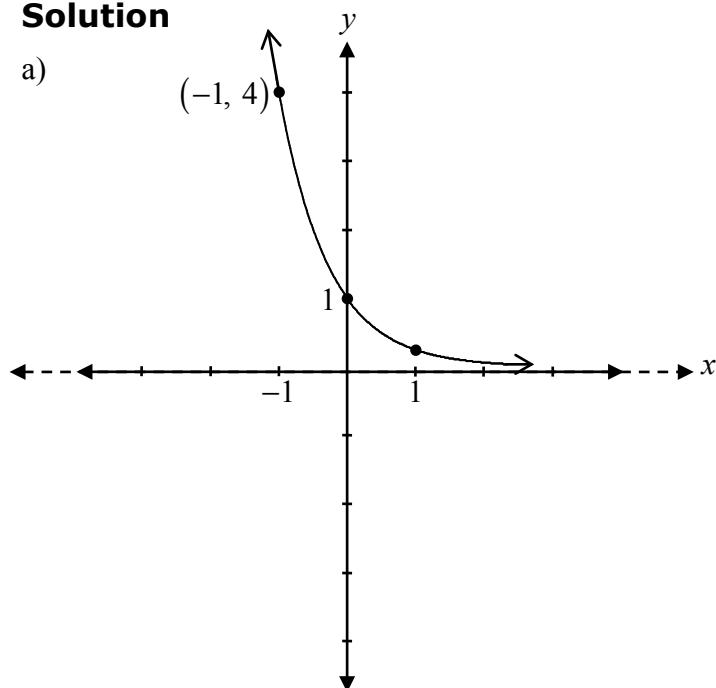
Sketch the graphs of:

a) $y = \left(\frac{1}{4}\right)^x$

b) $y = 2\left(\frac{1}{4}\right)^x$

Solution

a)



$\frac{1}{2}$ mark for decreasing exponential function

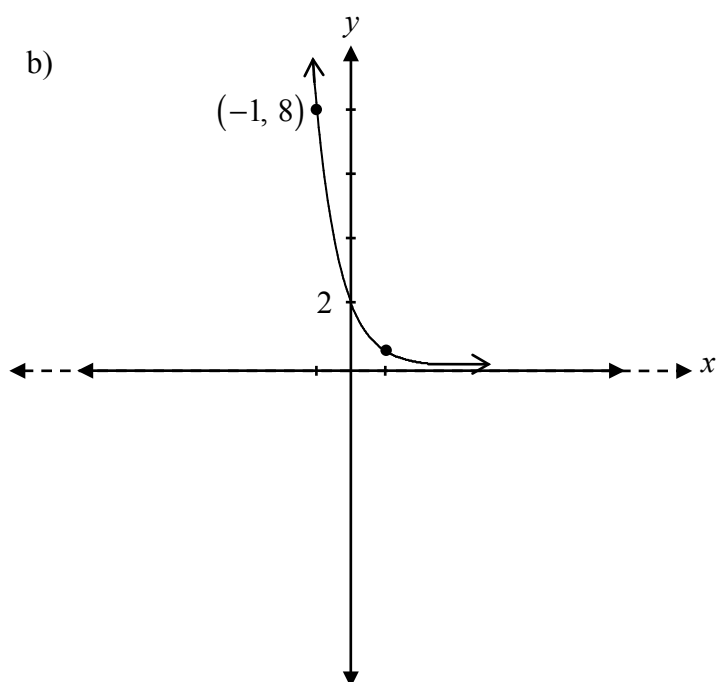
$\frac{1}{2}$ mark for y-intercept $(0, 1)$

$\frac{1}{2}$ mark for consistent point of an exponential function

$\frac{1}{2}$ mark for horizontal asymptote at $y = 0$

2 marks

b)



1 mark for a vertical stretch by a factor of 2 of the graph consistent with a)

1 mark

Determine all of the zeroes of the function $p(x) = x^3 - 5x^2 - 2x + 24$, given one of the factors of $p(x)$ is $(x - 3)$.

Solution

$$0 = x^3 - 5x^2 - 2x + 24$$

3	1	-5	-2	24
	↓	3	-6	-24
	1	-2	-8	0

½ mark for $x = 3$

1 mark for synthetic division (or for any equivalent strategy)

$$x^2 - 2x - 8 = 0$$

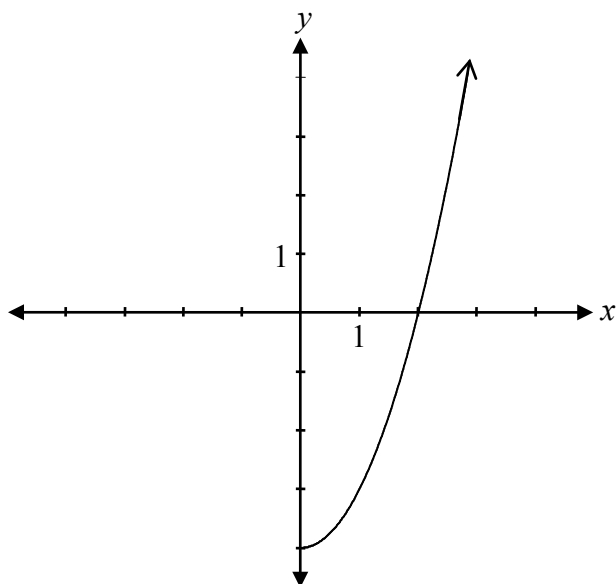
$$(x - 4)(x + 2) = 0$$

zeroes: 3, 4, -2

½ mark for consistent zeroes

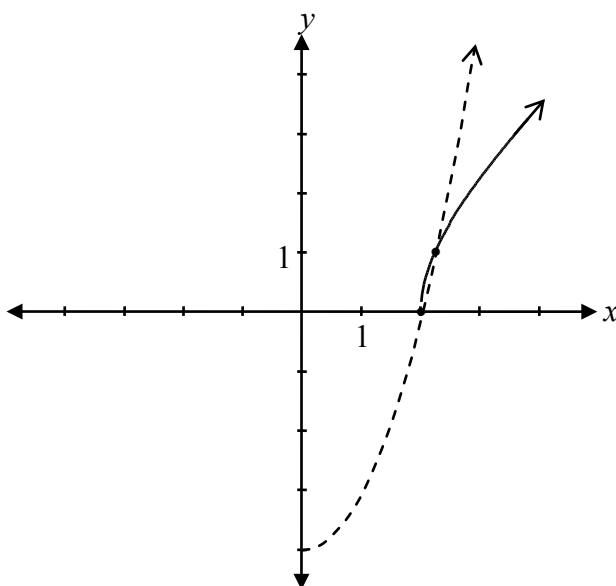
2 marks

Given the graph of $f(x)$,



sketch the graph of $y = \sqrt{f(x)}$.

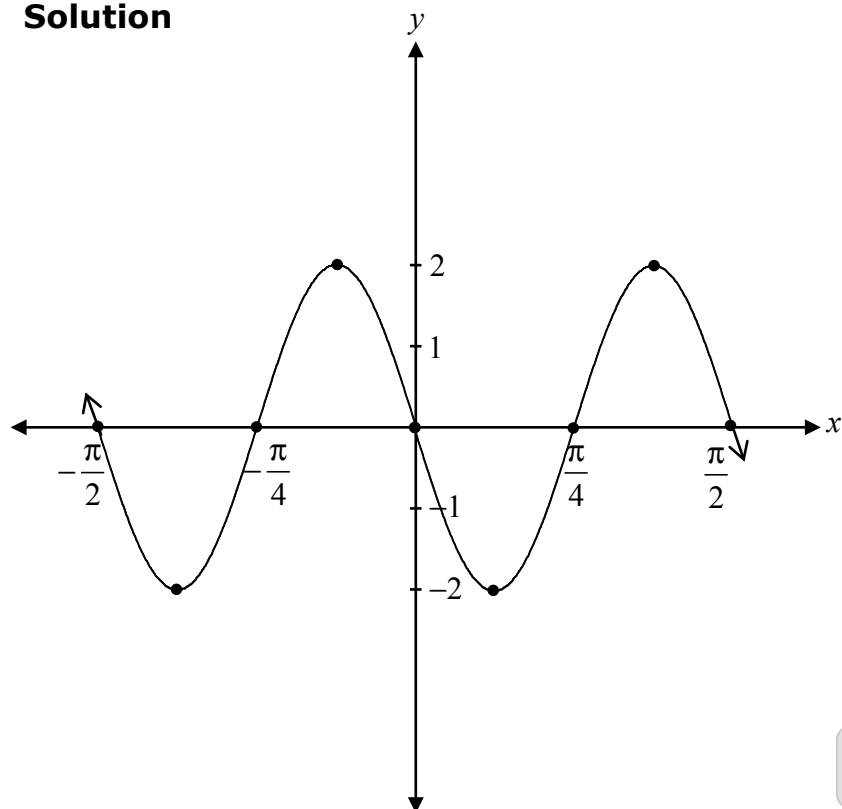
Solution



1 mark for restricting domain
 $\frac{1}{2}$ mark for shape between invariant points
 $\frac{1}{2}$ mark for shape to the right of the invariant points

2 marks

Sketch the graph of at least one period of the function $y = -2\sin(4x)$.

Solution

1 mark for amplitude
1 mark for period
1 mark for reflection in the x -axis

3 marks

Evaluate:

$$\frac{1}{2} \log_3 144 - \log_3 4 + 2 \log_3 3$$

Solution

$$\log_3 (144)^{\frac{1}{2}} - \log_3 4 + \log_3 (3)^2$$

1 mark for power rule

$$\log_3 12 - \log_3 4 + \log_3 9$$

$$\log_3 \left(\frac{12 \cdot 9}{4} \right)$$

$\frac{1}{2}$ mark for product rule

$\frac{1}{2}$ mark for quotient rule

$$\log_3 27$$

$$3$$

1 mark for evaluating a logarithm

3 marks

Match each function with its correct description.

- a) The graph of this function has a vertical asymptote at $x = -1$.
- b) The graph of this function has a point of discontinuity (hole) at $x = 3$.
- c) The graph of this function has a horizontal asymptote at $y = 4$.
- d) The domain of this function is $x \in \mathbb{R}$.

Place the appropriate letter in this column.

Solution

$$f(x) = \frac{4}{x^2 + 1} \quad \underline{\quad \text{d)} \quad}$$

$$g(x) = \frac{4x}{x + 3} \quad \underline{\quad \text{c)} \quad}$$

$$h(x) = \frac{4(x - 3)(x + 2)}{(x - 3)} \quad \underline{\quad \text{b)} \quad}$$

$$k(x) = \frac{4(x - 3)}{(x + 3)(x + 1)} \quad \underline{\quad \text{a)} \quad}$$

$\frac{1}{2}$ mark for each correct answer

2 marks

The point $(-3, 4)$ is on the graph of $y = \frac{1}{2}f(3x)$.

State the coordinates of the corresponding point on the graph of $y = f(x)$.

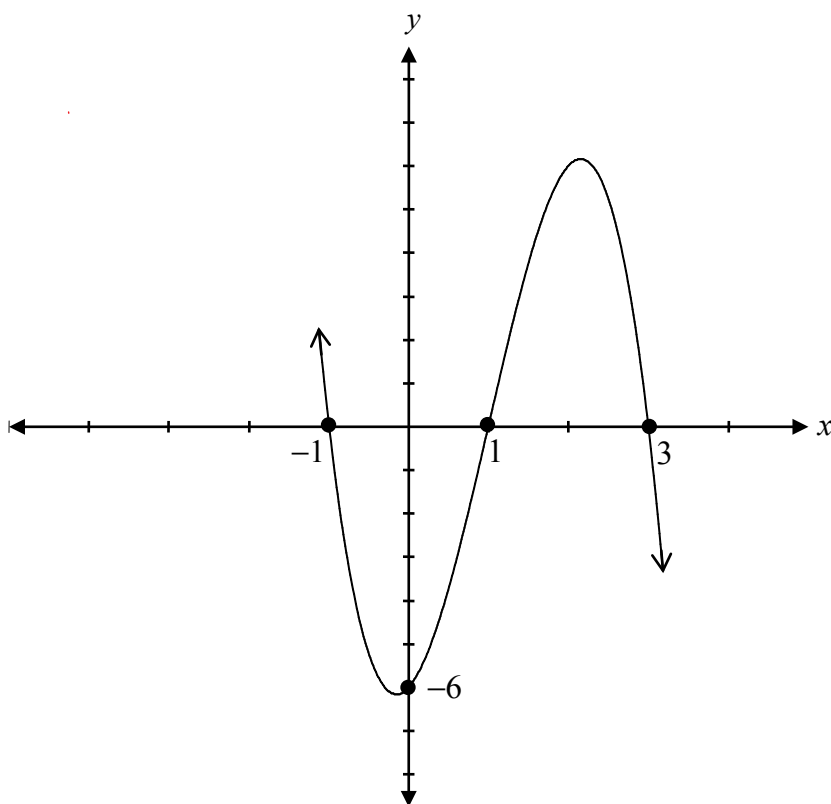
Solution

$(-9, 8)$

$\frac{1}{2}$ mark for each coordinate

1 mark

Sketch the graph of $y = -2(x-1)(x-3)(x+1)$.

Solution

1 mark for x -intercepts
1 mark for y -intercept
1 mark for end behaviour

3 marks

- a) Verify that the equation $\frac{1 - \sin^2 x}{\cos x} = \frac{\sin 2x}{2 \sin x}$ is true for $x = \frac{\pi}{3}$.
- b) Explain why verifying the equation for $x = \frac{\pi}{3}$ is insufficient to conclude that the equation is an identity.

Solution

$$\begin{array}{l|l}
 \text{a)} \quad \frac{1 - \sin^2\left(\frac{\pi}{3}\right)}{\cos \frac{\pi}{3}} & \frac{\sin 2\left(\frac{\pi}{3}\right)}{2 \sin \frac{\pi}{3}} \\
 \\
 \frac{1 - \left(\frac{\sqrt{3}}{2}\right)^2}{\frac{1}{2}} & \frac{\frac{\sqrt{3}}{2}}{2\left(\frac{\sqrt{3}}{2}\right)} \\
 \\
 \frac{1 - \frac{3}{4}}{\frac{1}{2}} & \frac{\frac{\sqrt{3}}{2}}{\frac{2}{\sqrt{3}}} \\
 \\
 \frac{\frac{1}{4}}{\frac{1}{2}} & \frac{1}{2} \\
 \frac{1}{2} &
 \end{array}$$

LHS = RHS

1 mark for exact values ($\frac{1}{2}$ mark for $\sin \frac{\pi}{3}$, $\frac{1}{2}$ mark for $\cos \frac{\pi}{3}$)1 mark for simplification ($\frac{1}{2}$ mark for LHS, $\frac{1}{2}$ mark for RHS)**2 marks**

- b) Proving that it is true for one value does not mean that it is true for all values.

1 mark

Evaluate:

$$\frac{{}_7P_2}{{}_7P_5}$$

Solution

$$\frac{\frac{7!}{(7-2)!}}{\frac{7!}{(7-5)!}}$$

½ mark for substitution

$$\frac{\frac{7!}{5!}}{\frac{7!}{2!}}$$

$$\frac{2!}{5!}$$

½ mark for simplification

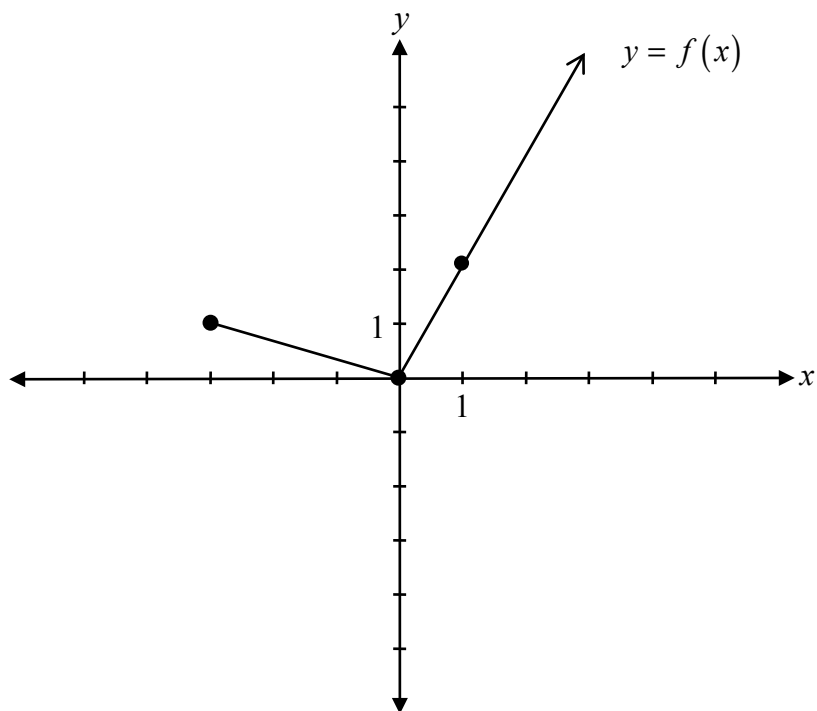
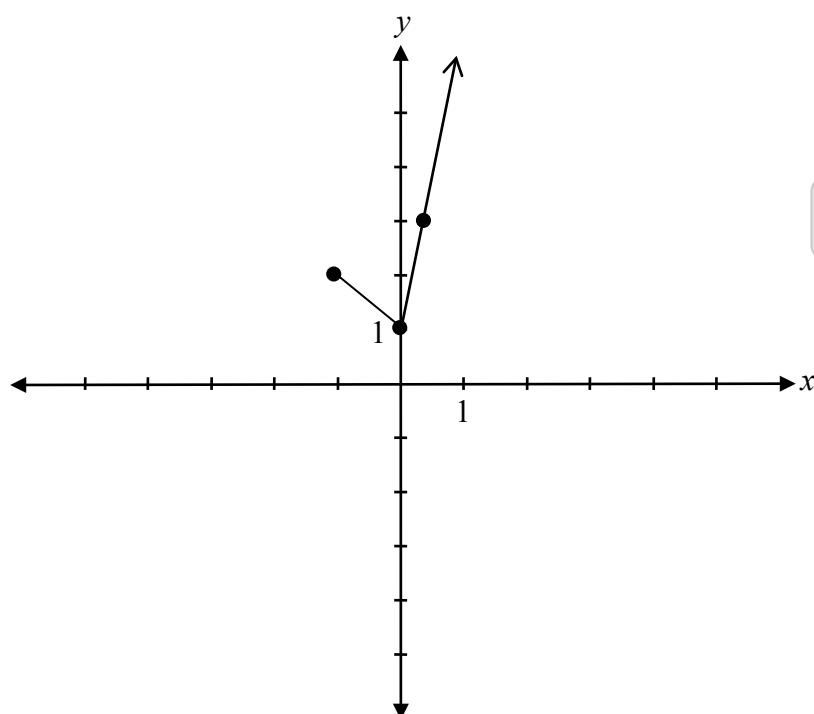
$$\frac{2 \times 1}{5 \times 4 \times 3 \times 2 \times 1}$$

1 mark for expanding factorials

$$\frac{1}{60}$$

2 marks

Use the graph of $y = f(x)$ to sketch the graph of $y = f(3x) + 1$.

**Solution**

1 mark for horizontal compression
1 mark for vertical translation

2 marks

Solve the following equation:

$$\log_4(x+2) + \log_4 3 = \log_4 x$$

Solution

Method 1

$$\log_4(x+2) + \log_4 3 = \log_4 x$$

$$\log_4[(x+2)3] = \log_4 x$$

$$3(x+2) = x$$

$$3x + 6 = x$$

$$x = -3$$

No solution

1 mark for product rule

1 mark for equating arguments

½ mark for solving for x

½ mark for rejecting extraneous root

3 marks

Method 2

$$\log_4(x+2) + \log_4 3 = \log_4 x$$

$$\log_4(x+2) + \log_4 3 - \log_4 x = 0$$

$$\log_4 \left[\frac{3(x+2)}{x} \right] = 0$$

$$4^0 = \frac{3x+6}{x}$$

$$x = -3$$

~~$$x = -3$$~~

1 mark for logarithmic rules (½ mark for product rule; ½ mark for quotient rule)

1 mark for exponential form

½ mark for solving for x

½ mark for rejecting extraneous root

3 marks

Determine the coordinates of the point of discontinuity (hole) for the graph of the function

$$y = \frac{(2-x)(x-3)}{(x-2)}.$$

Solution

$$x \neq 2$$

$$y = -(x-3)$$

$$y = -(2-3)$$

$$y = 1$$

$$(2, 1)$$

1 mark for point of discontinuity (hole) at $(2, 1)$

($\frac{1}{2}$ mark for $x = 2$, $\frac{1}{2}$ mark for consistent y -coordinate)

1 mark

Evaluate and simplify $\sec\left(\frac{5\pi}{6}\right) \cdot \tan\left(-\frac{\pi}{6}\right)$.

Solution

$$\left(-\frac{2}{\sqrt{3}}\right) \cdot \left(-\frac{1}{\sqrt{3}}\right)$$

$$\frac{2}{3}$$

1 mark for $\sec\left(\frac{5\pi}{6}\right)$ ($\frac{1}{2}$ mark for value, $\frac{1}{2}$ mark for quadrant)

1 mark for $\tan\left(-\frac{\pi}{6}\right)$ ($\frac{1}{2}$ mark for value, $\frac{1}{2}$ mark for quadrant)

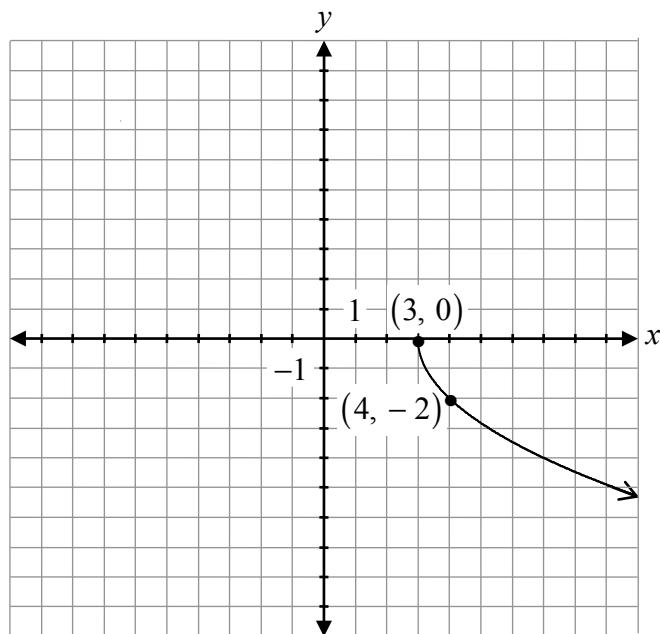
2 marks

Sketch the graph of the following function:

$$y = -2\sqrt{x-3}$$

Solution

Method 1



1 mark for shape (graph of a radical function)

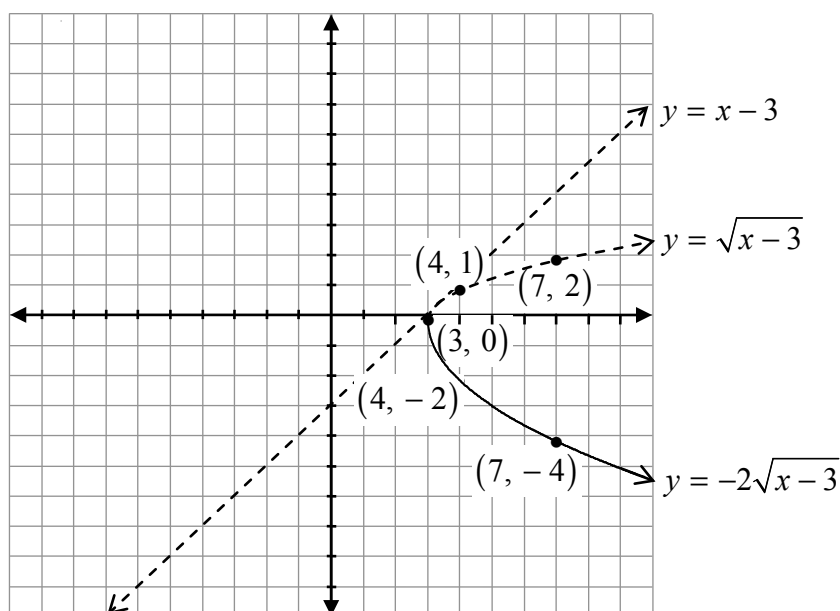
1 mark for vertical reflection

1 mark for horizontal shift

1 mark for vertical stretch

4 marks

Method 2



1 mark for invariant points where $y = 0$ and $y = 1$ ($\frac{1}{2}$ mark for each point)

1 mark for domain of $[3, \infty)$

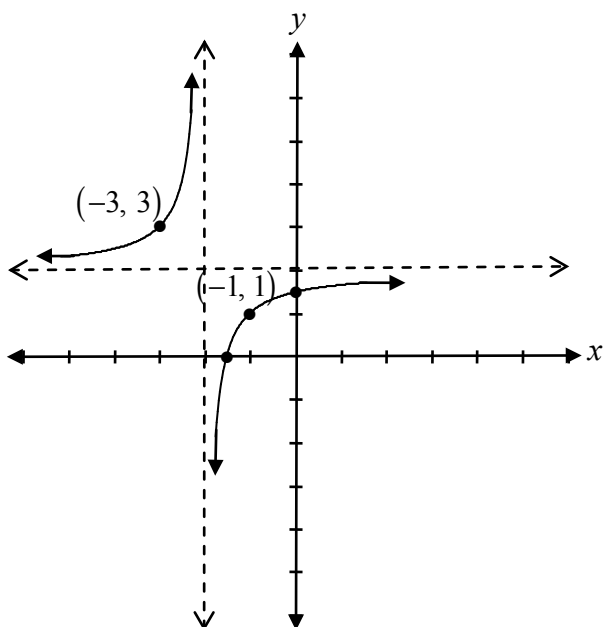
$\frac{1}{2}$ mark for shape between invariant points

$\frac{1}{2}$ mark for shape to the right of the invariant points

1 mark for applying transformations (vertical stretch, vertical reflection)

4 marks

Sketch the graph of $f(x) = \frac{2x+3}{x+2}$.

Solution

1 mark for vertical asymptote

1 mark for horizontal asymptote

$\frac{1}{2}$ mark for graph left of the vertical asymptote

$\frac{1}{2}$ mark for graph right of the vertical asymptote

3 marks

- a) Given the functions $f(x) = \sqrt{4+x}$ and $g(x) = |3x-6|$, evaluate $f(g(-5))$.
b) Is it possible to evaluate $g(f(-5))$?

Justify your answer.

Solution

a) $g(-5) = |3(-5) - 6|$

$$g(-5) = 21$$

1 mark for $g(-5)$

$$f(21) = \sqrt{4+21}$$

$$= \sqrt{25}$$

$$= 5$$

1 mark for consistent value of $f(g(-5))$

2 marks

- b) No, because $f(x)$ is undefined when $x = -5$
or

1 mark for justification

$$f(-5) = \sqrt{4+(-5)}$$

$$f(-5) = \sqrt{-1}$$

1 mark

$f(-5)$ is undefined because you cannot evaluate the square root of a negative number.

Identify which of these values is greater. Justify your answer.

$$\log_5 80 \quad \text{or} \quad \log_3 30$$

Solution

$$\begin{aligned} 5^2 &= 25 \\ 5^3 &= 125 \end{aligned} \quad \log_5 80 \text{ is less than } 3$$

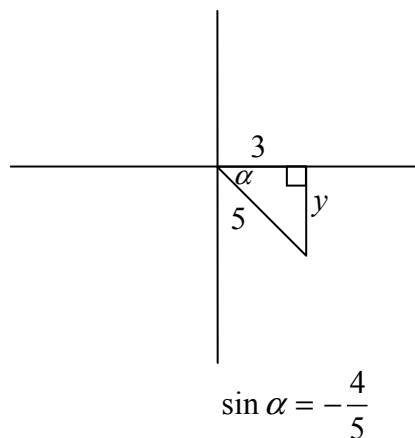
$$\begin{aligned} 3^3 &= 27 \\ 3^4 &= 81 \end{aligned} \quad \log_3 30 \text{ is more than } 3$$

$\therefore \log_3 30$ is greater

1 mark for justification

1 mark

Given $\cos \alpha = \frac{3}{5}$, where α is in quadrant IV, and $\cos \beta = -\frac{2}{3}$, where β is in quadrant II, determine the exact value of $\sin(\alpha - \beta)$.

Solution

$$x^2 + y^2 = r^2$$

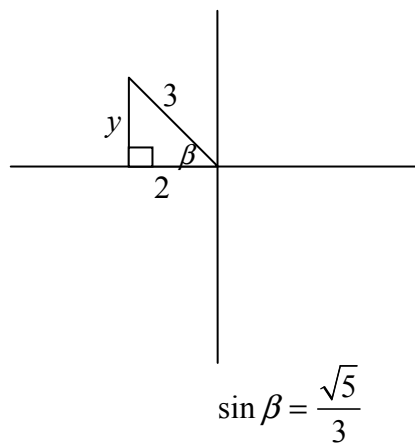
$$9 + y^2 = 25$$

$$y^2 = 16$$

$$y = \pm 4$$

$$y = -4$$

½ mark for value of y



$$x^2 + y^2 = r^2$$

$$4 + y^2 = 9$$

$$y^2 = 5$$

$$y = \pm\sqrt{5}$$

$$y = \sqrt{5}$$

½ mark for value of y

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$= \left(-\frac{4}{5}\right)\left(-\frac{2}{3}\right) - \left(\frac{3}{5}\right)\left(\frac{\sqrt{5}}{3}\right)$$

$$= \frac{8 - 3\sqrt{5}}{15}$$

½ mark for $\sin \alpha$

½ mark for $\sin \beta$

1 mark for substitution into correct identity

3 marks

Determine the number of possible sandwiches from the following menu.

MENU

Select one item from each column:

<u>Bread</u>	<u>Sauce</u>	<u>Meat</u>	<u>Vegetable</u>
White	Mayo	Turkey	Tomato
Rye	Mustard	Ham	Onion
Brown		Roast Beef	Lettuce
		Chicken	

Solution

$$3 \times 2 \times 4 \times 3$$

72 sandwiches

1 mark